

BlueMath-Hub: A cloud-based, open-source, python framework with interactive notebooks for statistical analysis and simulation of coastal climate hazards in a changing climate

Jared Ortiz-Angulo Cantos

Researcher, GeoOcean Group, Universidad de Cantabria (UC)

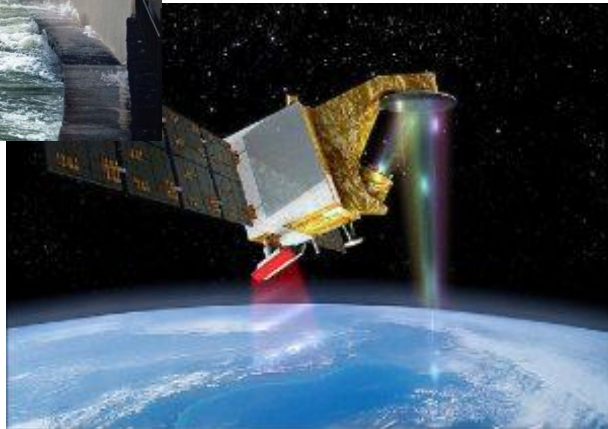
ProCoast 2025
October 20th-23th, 2025



What is BlueMath?



Coastal Hazards



Available tools



Environmental Modelling & Software

Volume 165, July 2023, 105712

CCdownscaling: A Python package for multivariable statistical climate model downscaling

Andrew D. Polasky^a

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<https://doi.org/10.1016>



Computers & Geosciences

Volume 162, May 2022, 105098

APPMAR 1.0: A Python application for downloading and analyzing of III® wave and wind data

German Rivillas-Ospina^a , Diego Casas^b, Mauro Antonio Marianella Bolívar^d, Gabriel Ruiz^e, Roberto Guerrero^d, José M. Hernández Milton Guerrero^c, Karina Díaz^d, Roberto del Río^c, Erick Campos^c

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<https://doi.org/10.1016/j.cageo.2022.105098>



CoastSeg: an accessible and extendable hub for satellite-derived-shoreline (SDS) detection and mapping

Sharon Fitzpatrick^a , Daniel Buscombe^a , Jonathan A. Warrick^a , Mark A. Lundine

^a Contracted to California, United States, California, Australia

DOI: [10.21105/joss.06683](https://doi.org/10.21105/joss.06683)



Environmental Modelling & Software

Volume 122, December 2019, 104528

CoastSat: A Google Earth Engine-enabled Python toolkit to extract shorelines from publicly available satellite imagery

Kilian Vos^a , Kristen D. Splinter, Mitchell D. Harley, Joshua A. Simmons, Ian L. Turner



Environmental Modelling & Software

Volume 178, July 2024, 106076

PyCoSMoS: An advanced toolbox for simulating real-world hydroclimatic data

Cappelli Francesco^a , Simon Michael Papalexiou^b , Yannis Markonis^c , Salvatore Grimaldi^a

Show more ▾

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Motivation

Only 11% of the
analyzed papers
provided code

Jansen et al., 2020

70% of the
researchers failed to
reproduce published
research

More than 50%
failed to reproduce
their own
experiments

Baker, 2016

Research progress slows down, as integrating **new methodologies**
into an existing framework requires significant **time and effort**

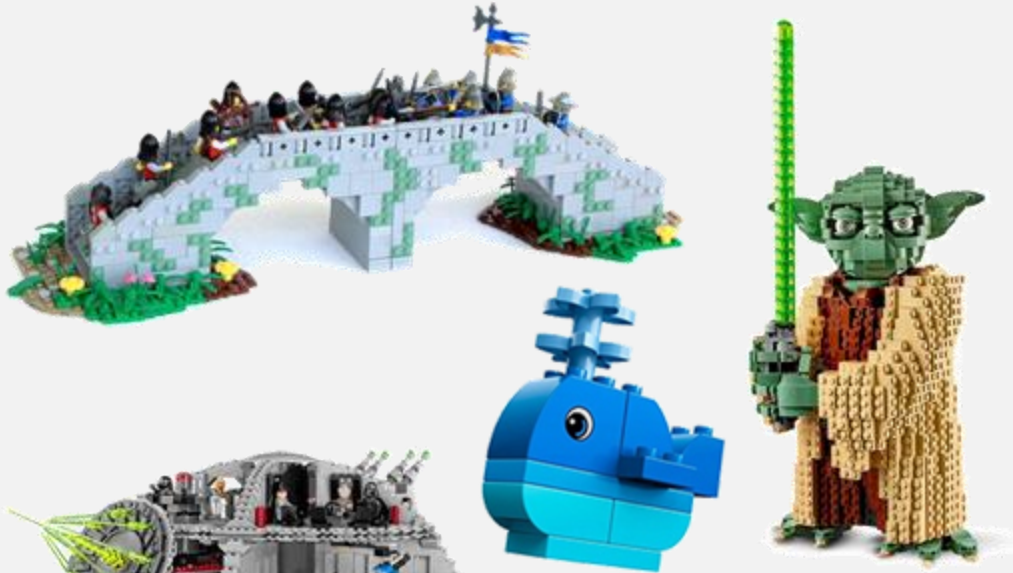
Projects and Publications



Projects and Publications

1

Identify individual pieces



Perez et al., 2015	Anderson et al., 2021
Cid et al., 2016	Cagigal et al., 2024
Rueda et al., 2017	van Vloten et al., 2023
Cagigal et al., 2019	Ricondo et al., 2023
Anderson et al., 2019	Ricondo et al., 2024



Projects and Publications



1

Identify individual pieces



Projects and Publications



2 Organize pieces



Projects and Publications

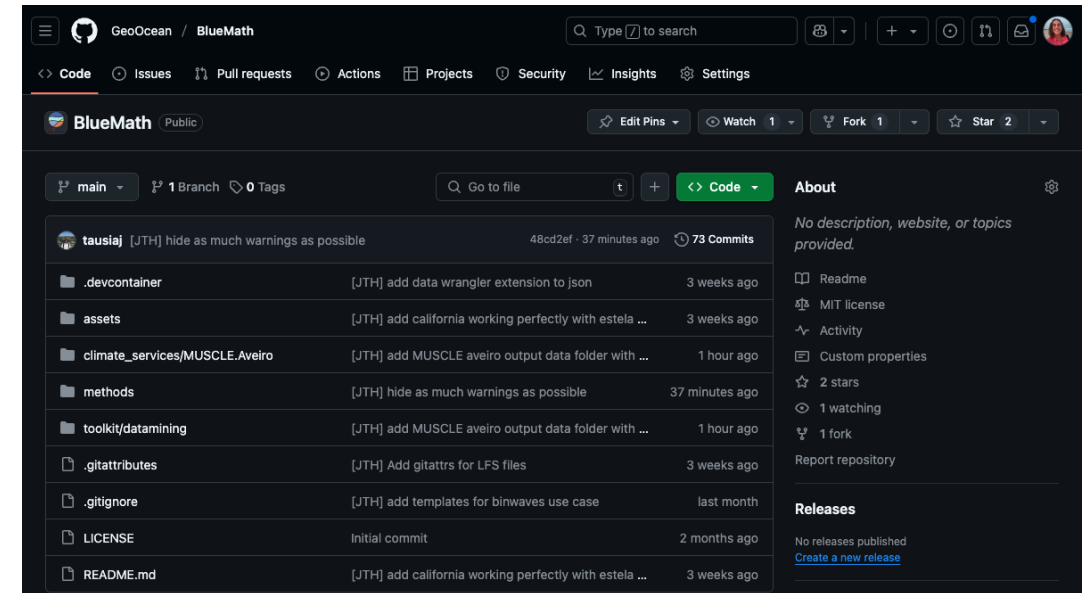


Perez et al., 2015
Anderson et al., 2021
Cid et al., 2016
Cagigal et al., 2024
Rueda et al., 2017
van Vloten et al., 2023
Cagigal et al., 2019
Ricondo et al., 2023
Anderson et al., 2019
Ricondo et al., 2024

3

Create an open-source code repository

<https://github.com/GeoOcean/BlueMath>



Structure



BlueMath

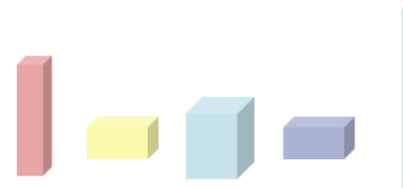
Structure



BlueMath



Toolkit

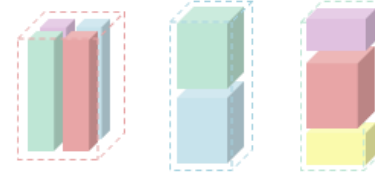




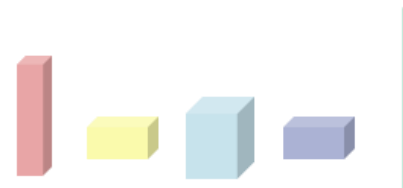
BlueMath



Methods



Toolkit

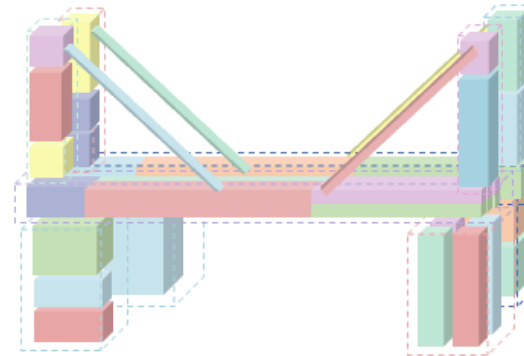




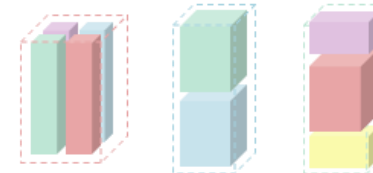
BlueMath



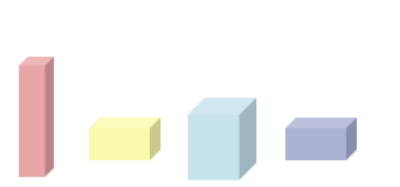
Climate Services



Methods



Toolkit

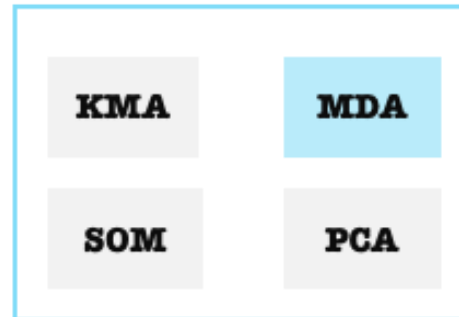




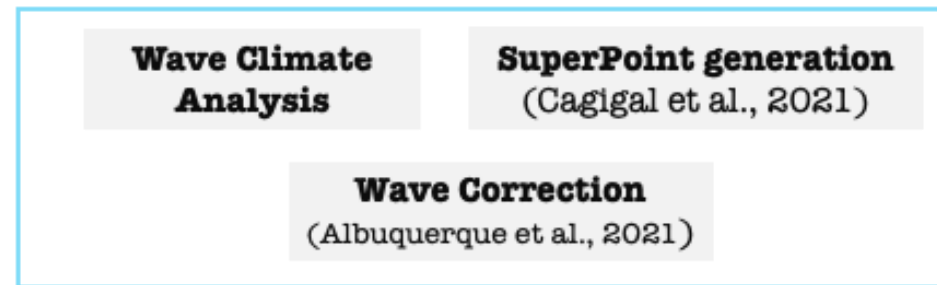
Toolkit



Datamining



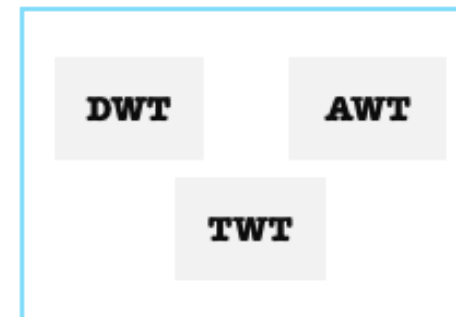
Waves



Wrappers



Predictor





Toolkit



MDA

Maximum Dissimilarity Algorithm

The screenshot displays a JupyterLab environment with a dark theme. The left sidebar shows a file explorer with a tree structure: **EXPLORADOR** > **BLUEMATH [CODESPACES: F...]** > **.devcontainer** > **assets** > **climate_services** > **methods** > **hybrid_downscaling** > **additive / BinWaves** > **outputs** > **templates** > **utils** > **BinWaves.Cantabria.Propagation...** (selected). Below this, other files like **BinWaves.Cantabria.Reconstruction.ip...** and **BinWaves.Cantabria.Validation.ipynb** are listed. The main area shows a Jupyter notebook titled **Comparison MDA vs KMeans**. The code in the notebook includes imports for **numpy**, **pandas**, **xarray**, **MDA**, **KMA**, **LHS**, and **plot_variable_combinations**. The **Load Dataset** section shows the loading of data from **/workspaces/BlueMath/climate_services/MUSCLE.Aveiro/data/era5_aveiro.nc**. The bottom status bar indicates the terminal is active, showing the prompt **[root@codespaces-3b176a BlueMath]#**.

```
import numpy as np
import pandas as pd
import xarray as xr
from bluemath.tk.datamining.mda import MDA
from bluemath.tk.datamining.kma import KMA
from bluemath.tk.datamining.lhs import LHS

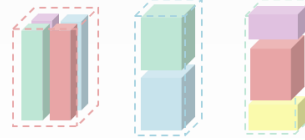
from utils.plotting import plot_variable_combinations
```

```
data = (
    xr.open_dataset("/workspaces/BlueMath/climate_services/MUSCLE.Aveiro/data/era5_aveiro.nc")
    .rename({"sw": "Hs", "pp1d": "Tp", "mwd": "Dir"})
    .to_dataframe()
    .iloc[:10, :]
    .reset_index()
    .drop(columns=["latitude", "longitude", "time"])
)
```

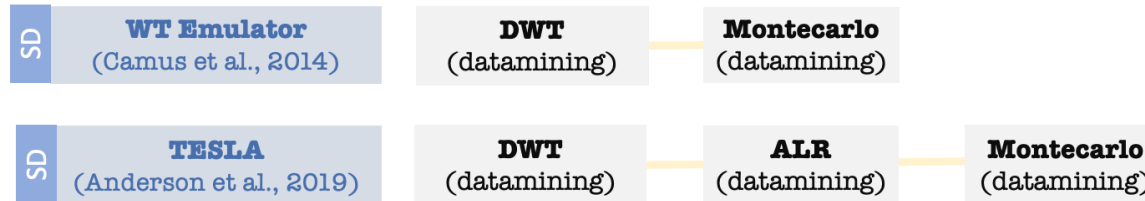
Methods



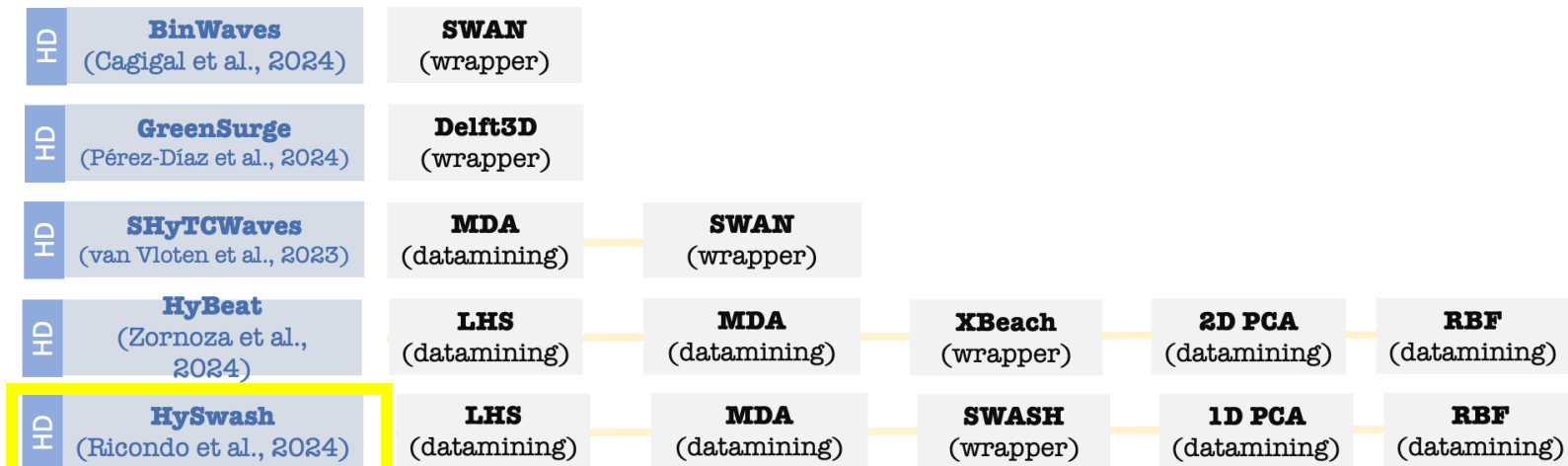
Methods

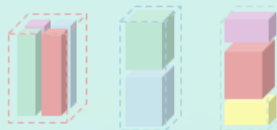


Statistical Downscaling



Hybrid Downscaling





HySwash

Maximum Dissimilarity Algorithm

BlueMath [Codespaces: fictional winner]

EXPLORADOR

- BLUEMATH [CODESPACES: FICTIONAL WINNER]
 - .devcontainer
 - assets
 - climate_services
 - methods
 - hybrid_downscaling
 - additive/BinWaves
 - metamodels/HySwash
 - assets
 - templates
 - utils
 - HySwash.Veggy.ipynb
 - HySwash.Veggy.RunUp.ipynb
 - statistical_downscaling
 - toolkit/datamining
 - utils
 - MDA_RBF.Aveiro.ipynb
 - MDA_RBF.ipynb
 - MDA_vs_KMeans.Aveiro.ipynb
 - MDA_vs_KMeans.ipynb
 - .gitattributes
 - .gitignore
 - LICENSE
 - README.md

HySwash: A hybrid method for nearshore wave processes

LHS
LATIN HYPERCUBE SAMPLING
Sampling Technique of Nearshore Wave Conditions (N cases) ($H_s, H_s/L_0, WL$)

MDA
MAXIMUM DISSIMILARITY ALGORITHM
Clustering Algorithm (selection of M cases)

SWASH
1D Numerical Simulation
 $R_{2\%}, \bar{\eta}, H_{rms}$
 H_{ss}, H_{ig}, H_{vlf}

from input to multiple spatial outputs

from input to singular output

PCA
PRINCIPAL COMPONENT ANALYSIS
Dimensionality Reduction

RBF
RADIAL BASIS FUNCTIONS
Interpolation Algorithm

Reconstruction of surf-zone hydrodynamics

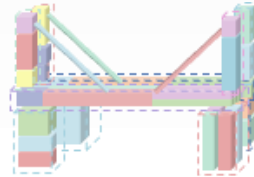
PROBLEMAS 8 SALIDA CONSOLA DE DEPURACIÓN TERMINAL PUERTOS 2 JUPYTER

[root@codespaces-3b176a BlueMath]#

Climate services



Climate Services



Wave Forecast System

 **Data Download**

Wave Correction
(Albuquerque et al., 2021)

SuperPoint generation
(Cagigal et al., 2021)

HD **BinWaves**
(Cagigal et al., 2024)

HD **HyBeat**
(Zornoza et al., 2024)

TC Forecast System

 **TC Data Download**

HD **SHyTCWaves**
(van Vloten et al., 2023)

SuperPoint generation
(Cagigal et al., 2021)

HD **BinWaves**
(Cagigal et al., 2024)

HD **HyBeat**
(Zornoza et al., 2024)

TC Risk Assessment

 **Data Download**

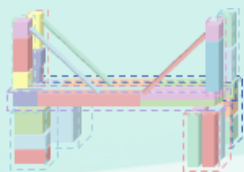
SD **TESLA**
(Anderson et al., 2019)

HD **SHyTCWaves**
(van Vloten et al., 2023)

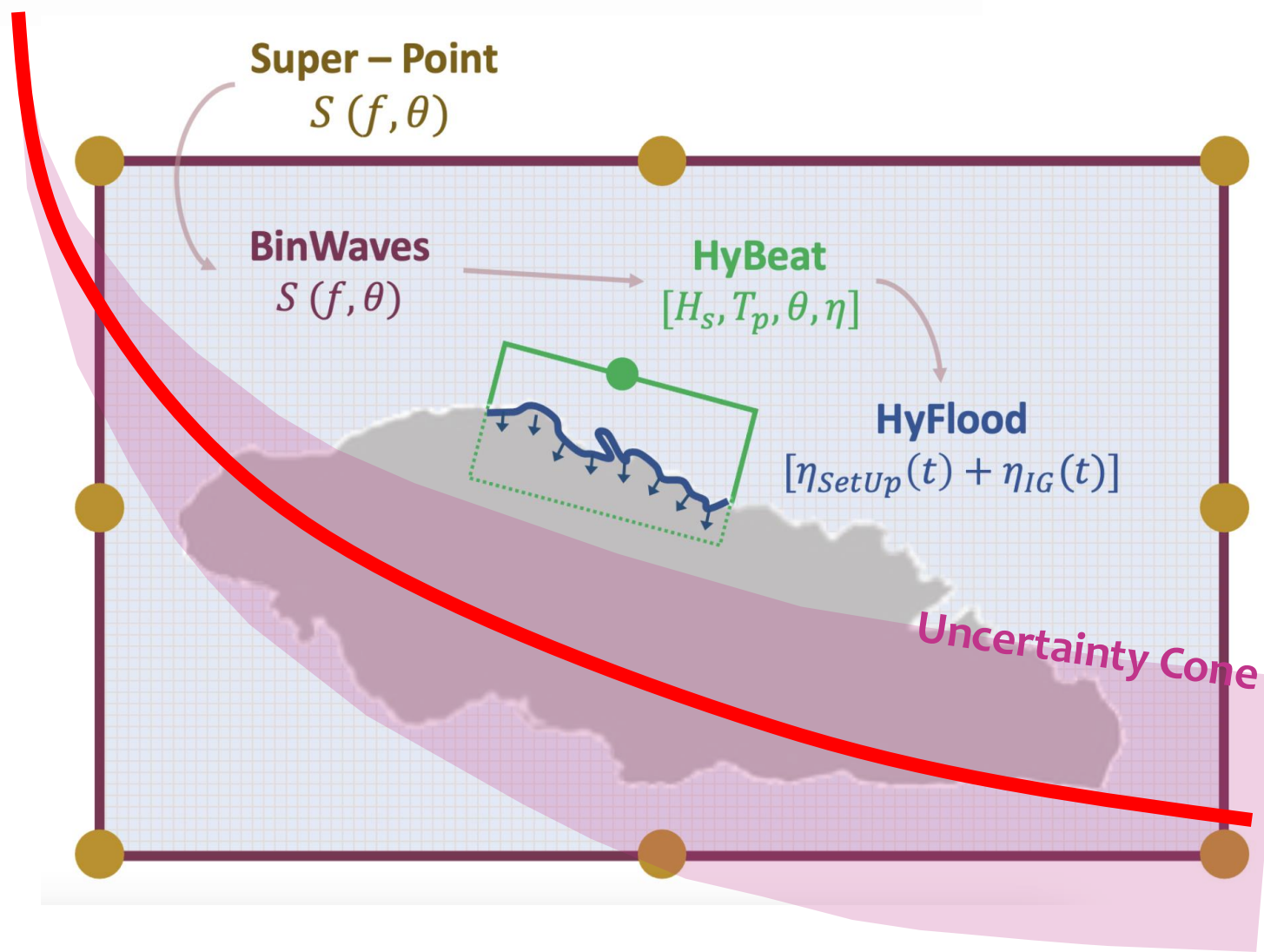
SuperPoint generation
(Cagigal et al., 2021)

HD **BinWaves**
(Cagigal et al., 2024)

HD **HyBeat**
(Zornoza et al., 2024)

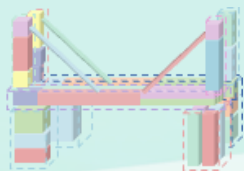


TC Forecast System





Climate
Services



UC

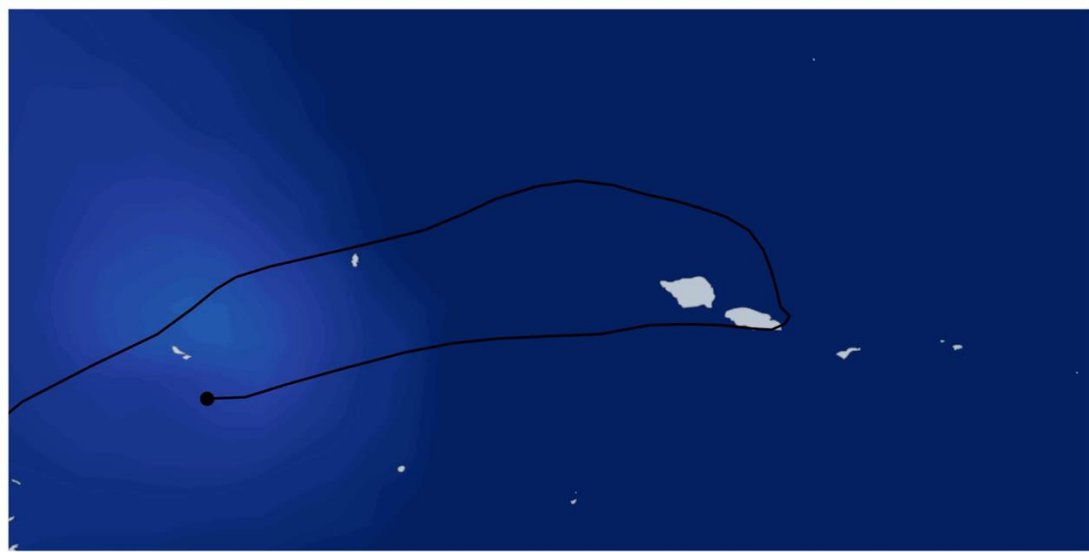
Universidad
de Cantabria



Geomatics and Ocean
Engineering Group
UNIVERSIDAD DE CANTABRIA

TC Forecast System

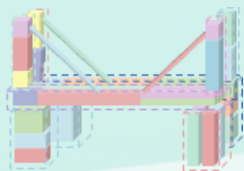
SHyTCWaves



van Vloten et al., 2024



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Services



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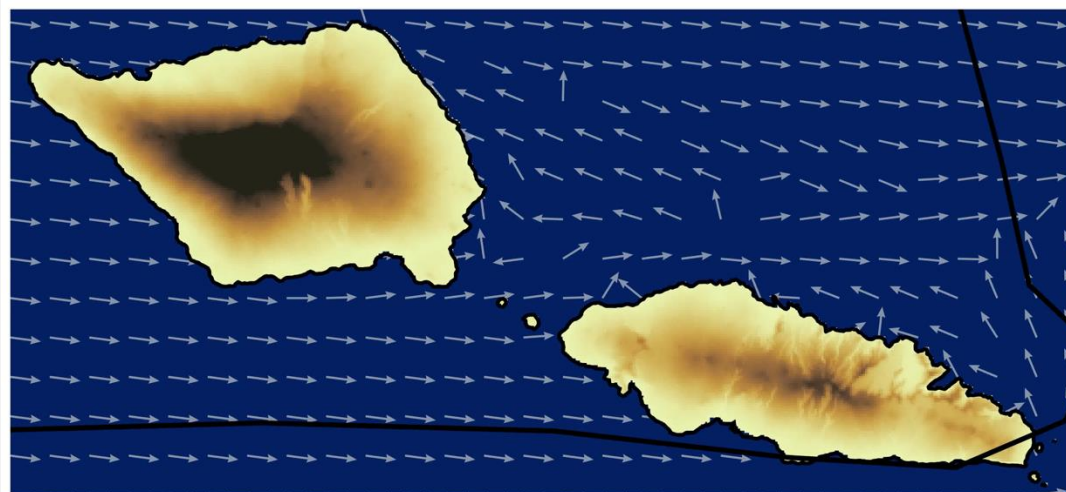
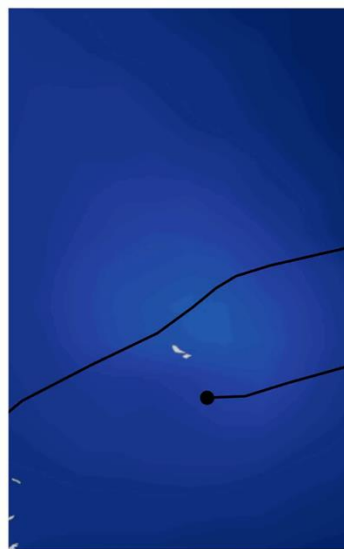


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TC Forecast System

SHyTCWaves

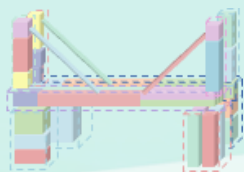
BinWaves



Cagigal et al., 2024



Climate
Services

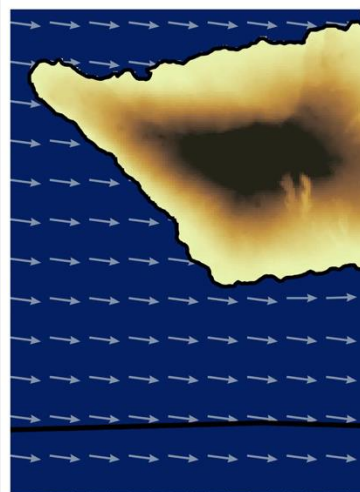
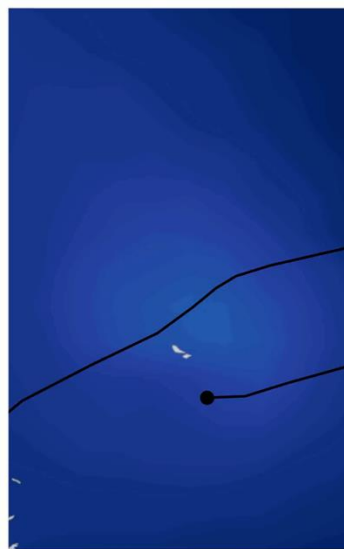


TC Forecast System

SHyTCWaves

BinWaves

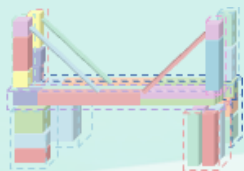
HyBeat



Zornoza et al., 2024



Climate
Services



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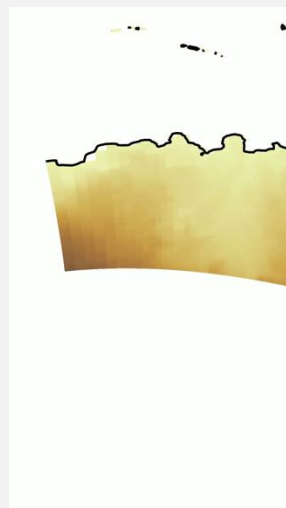
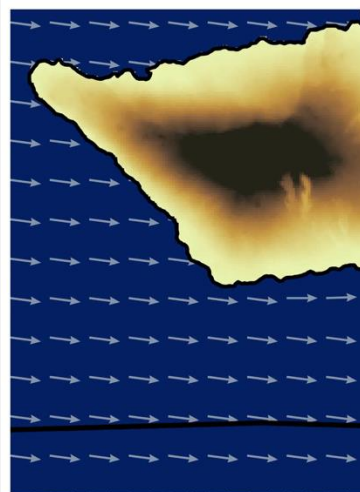
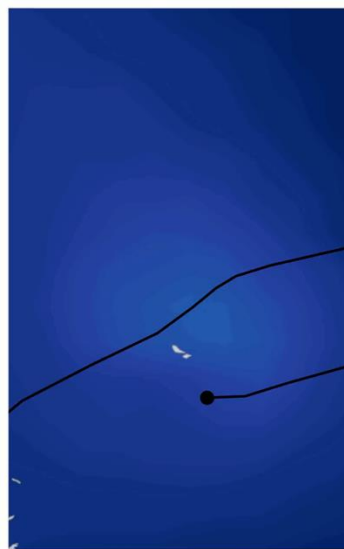
TC Forecast System

SHyTCWaves

BinWaves

HyBeat

HyFlood



How do we use / execute BlueMath?
How do we contribute to BlueMath?



BlueMath



Beginner



Curious



Skilled
user



Developer

GeoOcean / BlueMath

Type to search

Code

Issues 1

Pull requests

Actions

Projects

Security

Insights

Settings

BlueMath

Public

Edit Pins

Watch 3

Fork 7

Starred 17

develop

2 Branches

0 Tags

Go to file

Add file

Code

This branch is 14 commits ahead of main

tausiaj

Merge pull request #40 from CSAICC/develop

.devcontainer

[JTH] change devc

assets

[JTH] add svg logo

climate_services

[JTH] reduce resol

methods

upload README.m

toolkit

[AA] working versio

.gitattributes

[JTH] Add gitattr

.gitignore

[JTH] change names in hyswash and reformat templates ...

Local

Codespaces

Codespaces

Your workspaces in the cloud

On current branch

crispy acorn

develop* Uncommitted changes

On other branches

bug-free doodle

main* Uncommitted changes

Codespace usage for this repository is paid for by tausiaj.

About

No description, website, or topics provided.

Readme

GPL-3.0 license

Activity

Custom properties

17 stars

3 watching

7 forks

Report repository

Releases

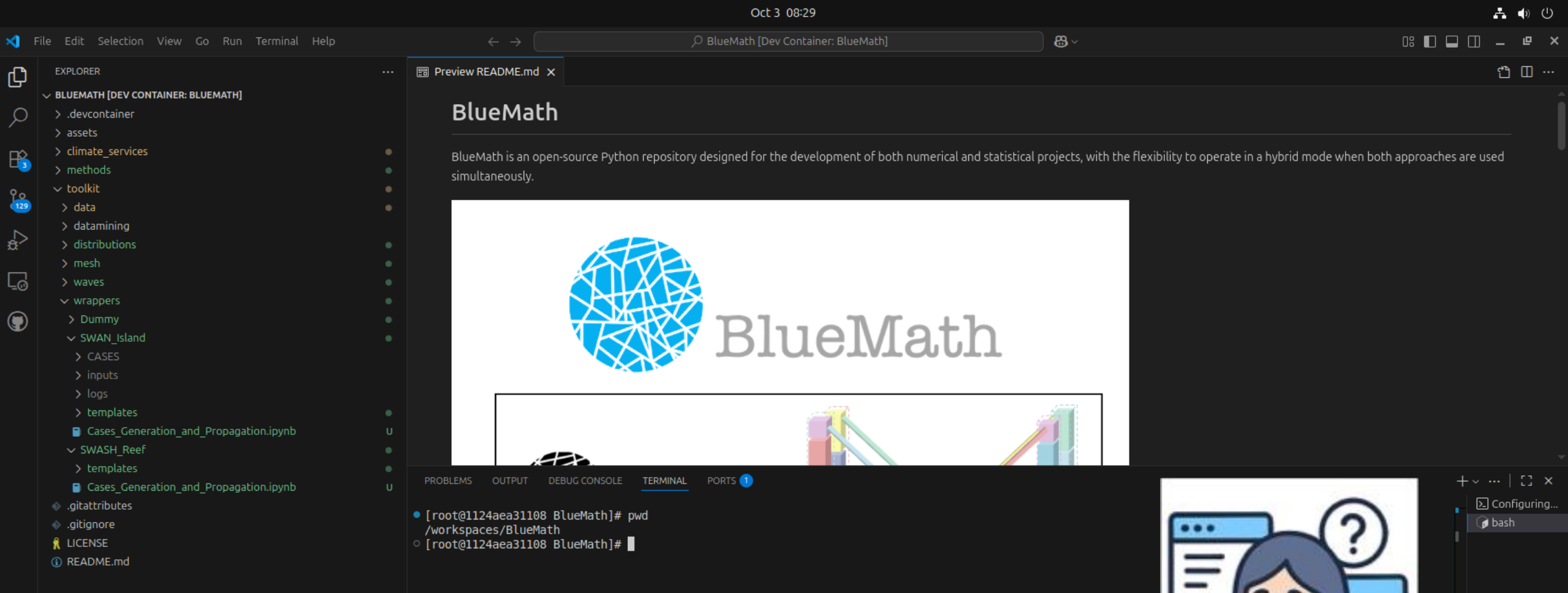
No releases published

Create a new release

Beginner

GitHub

GitHub Codespaces



Curious

1. Download **Visual Studio Code**



Curious

2. Download **miniconda**:

<https://www.anaconda.com/docs/getting-started/miniconda/install#macos-terminal-installer>

1. Download **git**: <https://git-scm.com/downloads/mac>

2. **Clone the git repo** in your local machine

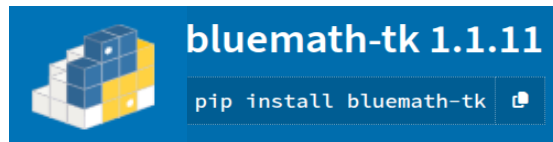
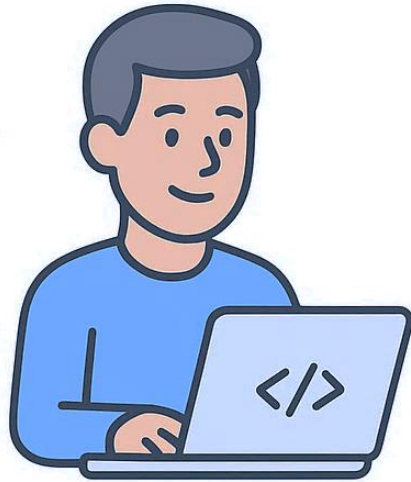
3. Change to the branch: **procast25**

4. Create a new **environemnt**: *conda create -n procoast25 python=3.11*

5. **Activate** the new environment: *conda activate procoast25*

6. Install **bluemath**: *pip install bluemath-tk*

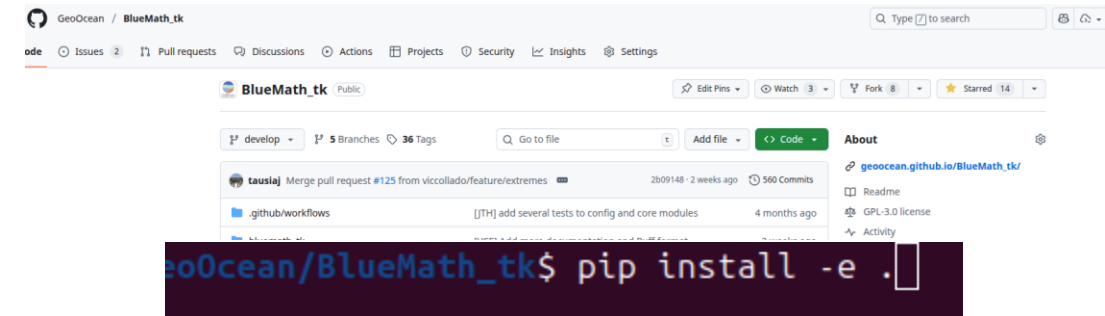
SKILLED USER



CONDA



DEVELOPER



BlueMath Funding

Internal project at the
GeoOcean group,
University of Cantabria



Specific Projects that partially support the development of BlueMath

CHANCE, Jose A.A. Antolinez (TU Delft)

BlueMath California, Borja G. Reguero (UCSC)

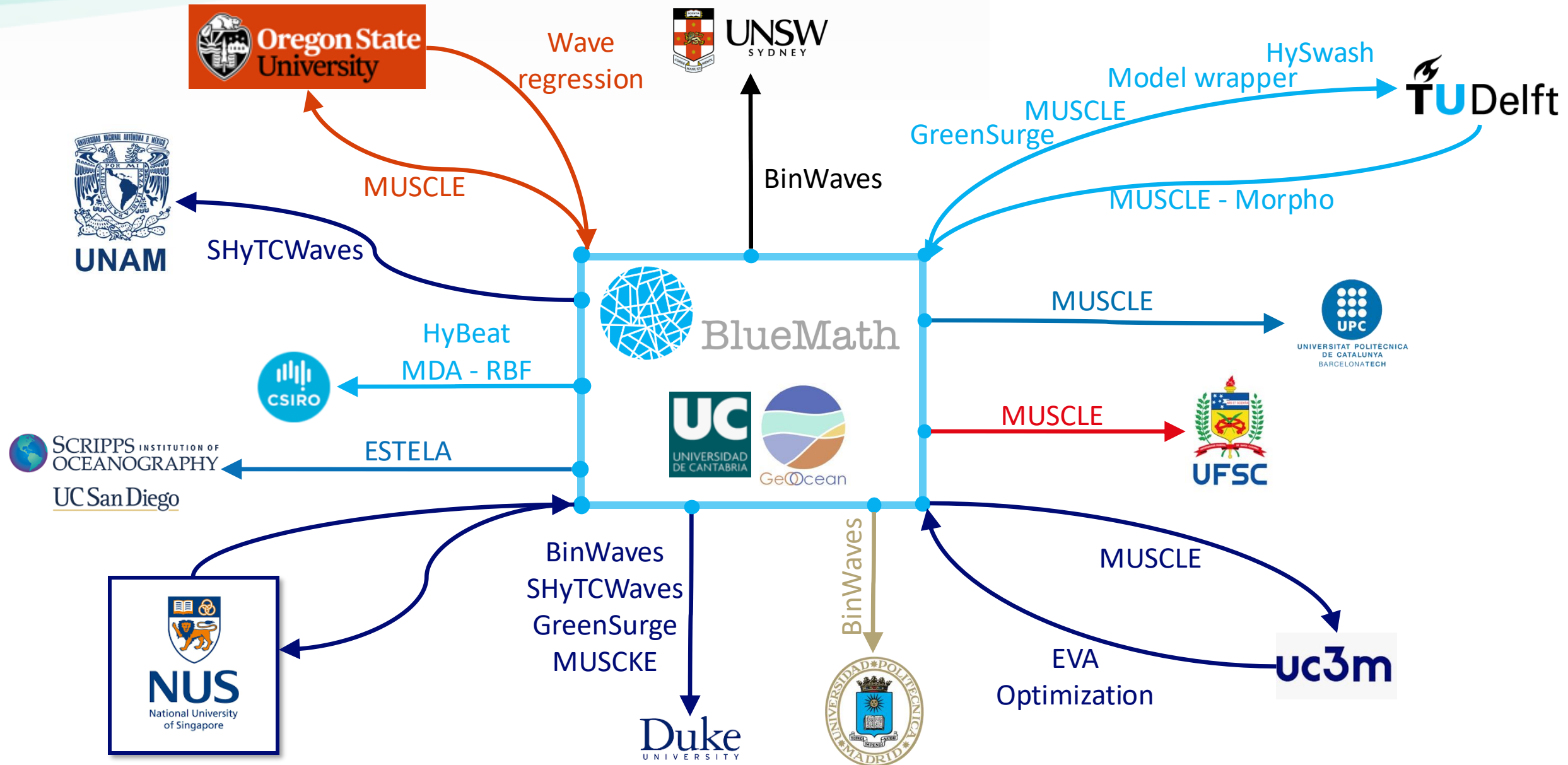
Australian BlueMath, Ron Hooke (CSIRO)

MURI, ESTCP: Peter Ruggiero (OSU)

Pacific climate indicators: John Marra (NOAA)



Users and code providers



Ongoing collaborations



Ongoing collaborations

Open for new collaborations

WAYS TO COLLABORATE:

- Developing Code
- Using and testing Code
- HPC Supercomputers
- BlueMath projects
- Hosting Courses



Courses



WDWE conference,
Hong Kong



Coastal Dynamics,
Aveiro, Portugal



41st IAHR World
Congress, Singapore



RCEM, Barcelona,
Spain

March 2025

June 2025

June 2025

October 2025

December 2024

April 2025

June 2025

September 2025

UFSC, Brazil



DigiShape,
Netherlands



NUS, Singapore



National Water Researches
Center – NWRC (Cairo,
Egypt)





javier.tausia@unican.es



Javier Tausia

Software Developer

Specialization

Software Development for Dynamic Coastal Systems

Bio

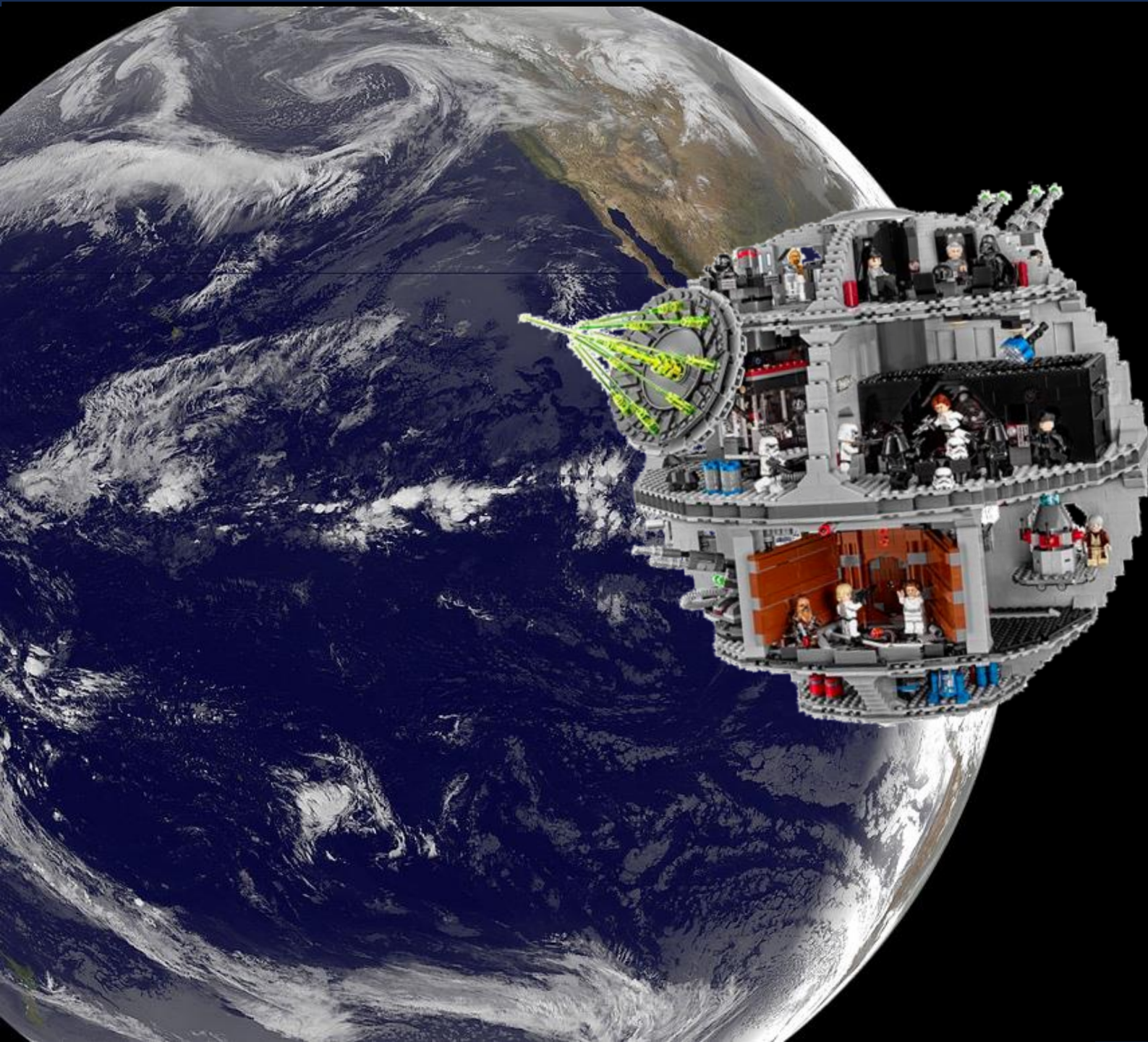
I studied my MSc in Data Science in the University of Cantabria (2020), where I also finished my Physics degree (2016). Since I finished my masters, I have been focused on the development of efficient meteo-oceanographic tools. Currently, I work as a technical researcher, exploring new and more efficient ways of coding that can improve its usability and scope in the group.

Education

MSc Data Science, University of Cantabria

Achievements

- Software Development Expert



Thank you
BlueMath Repo



Scan me